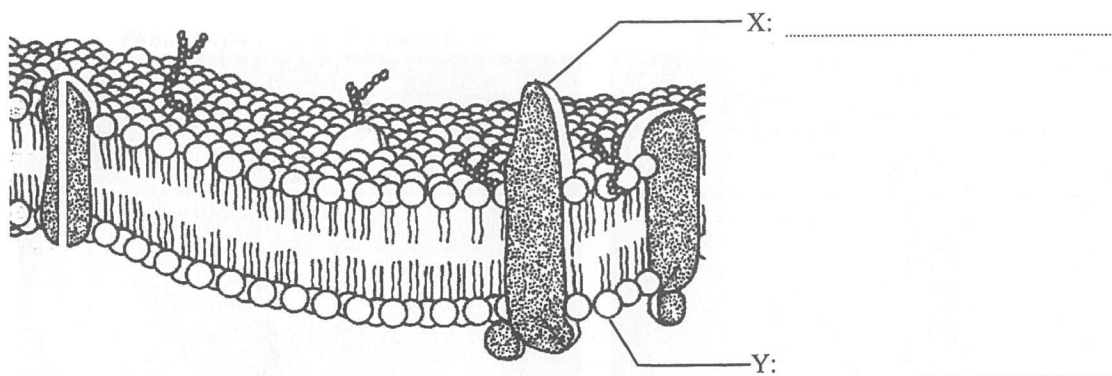
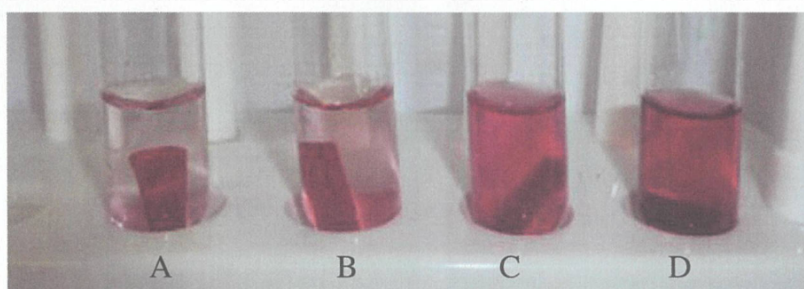


7. (a) The following is a schematic diagram of a cell membrane. In the spaces provided, label membrane components X and Y. (2 marks)



- (b) The vacuoles of beetroot cells contain a red pigment which will be released from the cells if the cell membrane and vacuole membrane are damaged. In an investigation, Gary placed identical cylinders of beetroot tissues into four test tubes. Each tube contained the same volume of alcohol at different concentrations. The following photograph shows the appearance of the solutions bathing the beetroot cylinders after 30 minutes:



- (i) From the result of the above investigation, deduce which test tube contained the highest concentration of the alcohol. (4 marks)

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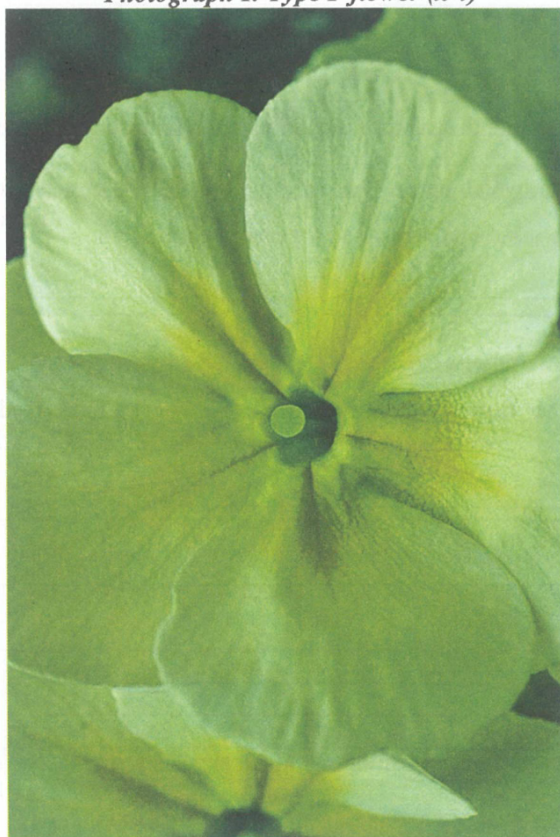
- (ii) After three hours, Gary found that the colour intensity of the solutions of all the test tubes became the same. Suggest an explanation for this. (2 marks)

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8. A primrose plant is a flowering plant that has two different types of flowers. Photograph I and Photograph II show the appearances of the two types of flowers (P and Q) and Photograph III and Photograph IV show the sections of the flowers respectively. Each primrose plant produces either type P flowers or type Q flowers.

Photograph I: Type P flower (x 4)



Photograph II: Type Q flower (x 4)



Photograph III: Section of type P flower (x 4)



Photograph IV: Section of type Q flower (x 4)



nectary

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- (a) Apart from the presence of nectaries, give *two* observable features of type P flowers which support the claim that the primrose is an insect-pollinated plant. (2 marks)

- (b) Butterflies collect nectar from flowers using a mouth structure in the form of a long sucking tube.

- (i) When a butterfly visits a type P flower, which part of the sucking tube will the pollen grains stick to? (1 mark)

- (ii) When the same butterfly visits another flower, which type of flower will be more readily pollinated? Why? (2 marks)

- (iii) What is the advantage of having the different positioning of anthers and stigmas in the primrose? (2 marks)

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9. Table I shows the number of deaths in Hong Kong caused by certain diseases. The deaths are categorised according to sex and age.

Table I

Cause of death	Sex	Age Group						
		All ages	0	1-4	5-14	15-44	45-64	≥65
Heart disease	Male	3352	1	2	2	104	679	2564
	Female	2981	2	2	0	20	146	2811
Diabetes mellitus	Male	213	0	0	0	2	43	168
	Female	246	0	0	0	10	19	217
Colon cancer	Male	725	0	0	0	16	177	532
	Female	627	0	0	0	8	131	488

- (a) Rank the diseases in the descending order of mortality (death rate). (1 mark)

- (b) From the data above, generalize a trend of mortality that is exhibited by all the diseases. (1 mark)

- (c) Table II shows the relative proportions of males and females in the above table who had particular lifestyles:

Table II

Lifestyle	Male	Female
Smoking daily	78.9%	21.1%
Never smoked	36.9%	63.1%
≥4 servings of processed meat per week	57.7%	42.3%

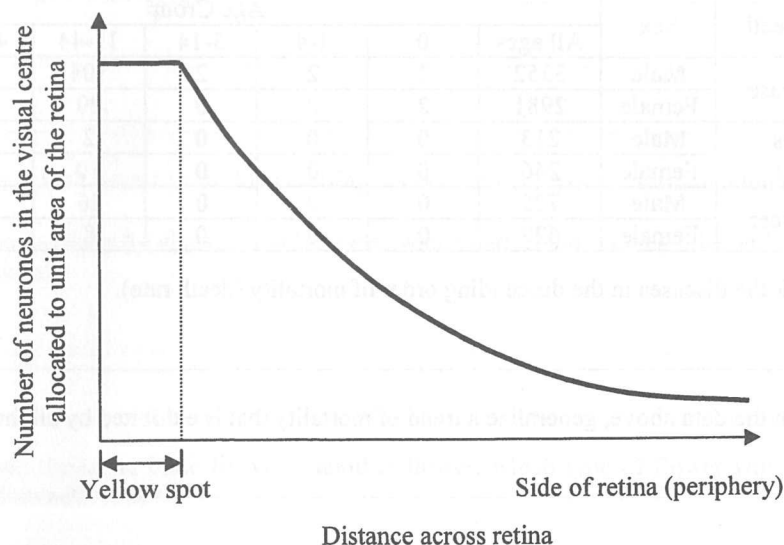
For the following diseases, explain how these lifestyles of the males and females are related to the number of deaths in Table I.

Heart disease: (3 marks)

Colon cancer: (3 marks)

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10. The visual centre of the brain is responsible for processing nerve impulses from the retina. The number of neurones in the visual centre allocated to receiving impulses from a unit area of the retina is shown below (only half of the retina is shown):



- (a) With reference to the types of photoreceptor cells and their distribution on the retina, explain why more neurones in the visual centre are allocated to unit area of the yellow spot. (4 marks)

- (b) After the perception of what we 'see', what it means to us depends on other parts of the brain. Explain how this works. (2 marks)

For the following question, candidates are required to present their answer in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

11. Recently, some people have adopted a diet rich in lean meat in order to lose weight and build muscle. They may be able to achieve these aims but there are some health problems associated with this diet. Discuss the pros and cons of this controversial diet with regard to the nutritional needs of our body and protein metabolism. (12 marks)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

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